```c
#include <stdio.h>

#define MAX 100

// Structure definition
struct Student {
    char name[50];
    int roll;
    float marks[3];
    float total;
    float percentage;
    char result[10];
};

int main() {
    struct Student s[MAX];
    int n, i, j;

    printf("Enter number of students: ");
    scanf("%d", &n);

    // Input student details
    for(i = 0; i < n; i++) {
        printf("\nEnter details for student %d\n", i + 1);

        printf("Name: ");
        scanf(" %[^\n]", s[i].name);

        printf("Roll Number: ");
        scanf("%d", &s[i].roll);

        s[i].total = 0;

        printf("Enter marks for 3 subjects:\n");
        for(j = 0; j < 3; j++) {
            printf("Subject %d: ", j + 1);
            scanf("%f", &s[i].marks[j]);
            s[i].total += s[i].marks[j];
        }

        s[i].percentage = s[i].total / 3;

        // Result calculation
        if(s[i].percentage >= 40)
            sprintf(s[i].result, "Pass");
        else
            sprintf(s[i].result, "Fail");
    }
```

```
    // Display results
    printf("\n--- Student Results ---\n");

    for(i = 0; i < n; i++) {
        printf("\nStudent %d\n", i + 1);
        printf("Name: %s\n", s[i].name);
        printf("Roll Number: %d\n", s[i].roll);
        printf("Total Marks: %.2f\n", s[i].total);
        printf("Percentage: %.2f%%\n", s[i].percentage);
        printf("Result: %s\n", s[i].result);
    }

    return 0;
}
```

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    }

    return 0;
}
```

```
computer@computer-Dell-Pro-Slim-Essential-QVS1260:~$ gcc home.c
computer@computer-Dell-Pro-Slim-Essential-QVS1260:~$ ./a.out
Enter number of students: 4

Enter details for student 1
Name: palak tulshan
Roll Number: 3
Enter marks for 3 subjects:
Subject 1: 99
Subject 2: 98
Subject 3: 97

Enter details for student 2
Name: sahil gujarathi
Roll Number: 2
Enter marks for 3 subjects:
Subject 1: 89
Subject 2: 90
Subject 3: 99

Enter details for student 3
Name: pushkar sable
Roll Number: 1
Enter marks for 3 subjects:
Subject 1: 89
Subject 2: 90
Subject 3: 89

Enter details for student 4
Name: aditya latke
Roll Number: 4
Enter marks for 3 subjects:
Subject 1: 99
Subject 2: 90
Subject 3: 67

--- Student Results ---

Student 1
Name: palak tulshan
Roll Number: 3
Total Marks: 294.00
Percentage: 98.00%
Result: Pass
```

```
Student 2
Name: sahil gujarathi
Roll Number: 2
Total Marks: 278.00
Percentage: 92.67%
Result: Pass

Student 3
Name: pushkar sable
Roll Number: 1
Total Marks: 268.00
Percentage: 89.33%
Result: Pass

Student 4
Name: aditya latke
Roll Number: 4
Total Marks: 256.00
Percentage: 85.33%
Result: Pass
```